Whiteboard Alive: Draw your own diagram

Draw a physics diagram with the **thick black marker**, we take a photo, and **it turns into a live simulation**.

1 Pick one of these

DIAGRAM	DRAW THIS	IT WILL...
Pendulum	A ceiling line, a string hanging from it, a circle at the end	**swing**
Spring	A ceiling (or wall) line, a zig-zag, a box at the end	**bounce**
Incline	A triangle, with a box sitting on the slope	**slide down**
Rolling ball	A triangle, with a circle sitting on the slope	**roll down**
Projectile	A ground line, a ball, and an arrow (or a cannon) pointing where it goes	**fly in an arc**
Pulley	A circle near the top, a line down each side, a box on each line (make one bigger)	**the heavier side drops**
Collision	A floor line, two circles on it, and an arrow above each one showing which way it moves	**crash and bounce**
Seesaw	A small triangle, a long plank on top of it, a box on each end (make one bigger)	**tip toward the heavier side**

Anything else? Draw it anyway. If we can't tell what it is, it still falls.

2 How to draw it so it works

- **Draw big**: the diagram should fill at least half of the paper.
- **Use the thick black marker only.** The app works best with dark black lines; pencil and thin pens may not be picked up. Press firmly.
- **Keep lines connected**: the string touches the ceiling, the zig-zag touches the box.
- **Keep words and numbers away from the parts**, about one finger's width, so they don't get mixed up with the lines.

You don't need to write any numbers. Values such as string length and mass can be changed by tapping them right under the moving picture.

3 Taking the photo

- Hold the phone **straight above** the paper.
- Make sure the **whole sheet** is in the shot.
- Avoid casting the phone's shadow over the drawing.

Then watch your drawing turn into clean shapes and start moving. You can also change gravity with the slider, tap **Moon**, or tap **Talk** and say *"make gravity weaker"*.